University of Florida, Department of Mathematics

Numerical Analysis PhD Exam, January 2002

Do any 8 of the following 10 problems. Problems 1–5 are Part 1: Numerical Linear Algebra. Problems 6–10 are Part II: Numerical Analysis.

1. This problem has three parts.

(a) Prove: If $A \in \mathbb{R}^{n\times n}$ has a set of n linearly independent eigenvectors, then A can be diagonalized.

(b) Prove: If $A \in \mathbb{R}^{n\times n}$ is symmetric, then A has an orthogonal set of eigenvectors.

(c) Diagonalize the matrix

$$A = \begin{pmatrix} 66 & 12 \\ 12 & 59 \end{pmatrix}.$$

2. This problem has three parts.

(a) Is the matrix

$$A = \begin{pmatrix} 5 & 1 \\ 0 & 3 \end{pmatrix}$$

diagonalizable? Why?

(b) Can the above matrix A be diagonalized by an orthogonal matrix S?

(c) State a theorem or condition that ensures a matrix can be diagonalized by an orthogonal matrix.

3. This problem has three parts.

(a) Give a careful statement of the singular value decomposition.

(b) Give a careful proof of the singular value decomposition.

(c) Compute the SVD for the matrix

$$A = \begin{pmatrix} -6 & 17 \\ 18 & -1 \end{pmatrix}.$$

4. Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Determine the orthogonal projection of A onto the column space of A.

(b) Compute the QR factorization of A.

5. This problem has five parts.

(a) Define the operator 2-norm for any matrix A in $\mathbb{R}^{m\times n}$.

(b) Prove: If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times q}$, then $\|BA\|_2 \le \|B\|_2 \cdot \|A\|_2$.

(c) Prove: If A is any matrix in $\mathbb{R}^{n\times n}$ and $\|A\|_2 < 1$, then $\lim_{n\to\infty} A^n = 0$.

(d) Prove: If A is a matrix in $\mathbb{R}^{n\times n}$, then define e^A as a power series expansion. Show this series converges.

(e) Prove: If $A \in \mathbb{R}^{n\times n}$ is invertible, b and Δb are vectors in $\mathbb{R}^n$, x and Δx are vectors in $\mathbb{R}^n$ and are solutions to $Ax = b$ and $A\Delta x = \Delta b$, respectively, then

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \kappa_2(A)\frac{\|\Delta b\|_2}{\|b\|_2}.$$

6. This problem has three parts.

(a) State Newton's method (the algorithm) for solving $f(x) = 0$ where $f : \mathbb{R} \to \mathbb{R}$.

(b) Assuming f is smooth and we have an initial guess sufficiently close to the root, state sufficient condition(s) for the method to converge quadratically.

(c) Prove: Let $T(x) : [a, b] \to [a, b]$ be a function such that $T'''(x)$ is continuous for all $x \in [a, b]$, $T(p) = p$, $T'(p) = T''(p) = 0$. If $x_0 \in [a, b]$ is a point with the property that the sequence defined recursively by $x_{k+1} = T(x_k)$ converges to p, then the sequence $\{x_k\}_{k=1}^{\infty}$ converges cubically to p.

7. This problem has four parts.

(a) Give a careful statement of the minimization property for clamped cubic splines.

(b) Give a careful statement of the convergence theorem for clamped cubic splines.

(c) Give a careful statement of the convergence theorem for the second derivatives for clamped cubic splines.

(d) Outline a proof of the convergence theorem for clamped cubic splines.

8. This problem has two parts.

(a) Given a set of data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ in the plane, describe how to find the conic section of the form $Ax^2 + Bxy + Cy^2 + Dx + Ey + 1 = 0$ which best fits the data in the sense of least squares.

(b) How do you determine whether or not the conic section is an ellipse, a hyperbola, or a parabola?

9. Triple Recursion Formula. If $\{\phi_n(x)\}$ is an orthogonal family of polynomials on $[a, b]$, with respect to the weight function $w(x) \geq 0$, and $n \geq 1$, then show that $\phi_{n+1}(x) = (a_n x + b_n)\phi_n(x) - c_n\phi_{n-1}(x)$, for some constants a_n, b_n, and c_n. (Hint: Let $g(x) = \phi_{n+1}(x) - a_n x\phi_n(x)$, where a_n is a constant chosen so that $g(x)$ has degree n.)

10. This problem has four parts.

(a) Give a careful statement of the contraction mapping theorem.

(b) Prove the contraction mapping theorem.

(c) Determine whether or not the function $T(x) = \frac{1}{4}\sin(3x + 2) + 12$ is a contraction.

(d) Discuss the convergence rate ensured by the contraction mapping theorem. (ie Is it linear, quadratic, or cubic?)